

Rank	Team	MAE	Rank	Team	MAE
1	TM-TREK	15.68	16	mahaalblooki	25.95
2	Alpom	15.94	17	RUG-5	26.07
3	USTC-BUPT	17.70	18	mahsaamani	26.27
4	ywnh111	18.08	19	aktsvigun	26.40
5	ywnh222	18.51	20	skillissue	27.99
6	Rkadiyala	18.54	21	NootNoot	28.01
7	DeepPavlov	19.25	22	TueCICL	34.88
8	knk42	19.42	23	dandread	35.17
9	vasko	19.93	24	novice8	44.82
10	logiczmaksimka	19.93	25	jelaron	48.14
11	AISPACE	21.19	26	TueSents	58.95
*	baseline	21.54	27	MasonTigers	60.78
12	ericmxf	21.55	28	Unibuc - NLP	74.28
13	blain	21.80	29	lanilequiu	78.18
14	1024m	22.36	30	scalar	87.72
15	cmy99	24.68			

Table 12: **Subtask C: Boundary Identification.**

score the importance of leveraging advanced LLMs, ensemble techniques, and comprehensive analysis to achieve superior performance in detecting machine-generated text across multilingual contexts.

6.2 Subtask B

For Subtask B (Multi-Way detection), 70 teams participated, with 20 surpassing the baseline of 74.61 accuracy. Table 11 displays the full results. In summary, the subtask results underline the effectiveness of diverse and innovative approaches, including fine-tuning advanced models (e.g., RoBERTa, DeBERTa, XLNet, Longformer, T5), data augmentation (e.g., using Subtask A instances), ensemble strategies, and the exploration of novel loss functions and learning techniques. The leading entries showcased a range of methodologies, from leveraging the power of large language models and addressing embedding anisotropy to integrating traditional and neural methods, underscoring the dynamic and evolving nature of NLP research. For instance, Team AISPACE utilized a weighted Cross-Entropy loss and an ensemble approach based on model performance per class, which led to the highest accuracy of 90.85.

6.3 Subtask C

Of the 30 systems that were submitted for Subtask C, 11 outperformed the baseline MAE of 21.54. The top system, TM-TREK, achieved the best submitted MAE of 15.68. A significant majority of the top-performing teams relied on ensembles of PLMs, indicating a consensus that combining the strengths of multiple models can lead to more robust and accurate predictions. This approach lever-

ages the diverse representations and strengths of different models to mitigate weaknesses inherent in individual systems.

Data augmentation emerged as a critical strategy among leading teams, suggesting its effectiveness in enhancing model performance by providing a richer, more varied training dataset. This includes both the generation of new training examples and the manipulation of existing data to better capture the complexity and variability of natural language.

Despite the advanced methodologies deployed, some teams struggled with issues related to overfitting and the adequacy of word embeddings. This underscores the ongoing challenges in developing models that generalize well to unseen data and the critical role of embeddings in capturing semantic and syntactic nuances of language.

7 Conclusion and Future Work

We have described SemEval-2024 Task 8 on Multi-generator, Multidomain, and Multilingual Machine-Generated Text Detection. The task garnered significant interest from researchers, with 126, 59, 70, and 30 teams submitting entries for Subtask A Monolingual, Subtask A Multilingual, Subtask B, and Subtask C, respectively. Additionally, we received 54 system description papers before finalizing this submission.

Overall, Subtasks A and B were relatively easier, with all systems showing improvements over the baseline. However, Subtask C proved to be significantly more challenging. Fewer teams participated, and many struggled to surpass our baseline results set in (Wang et al., 2024a).

In future work, we plan to extend our focus beyond machine-generated text detection to other modalities such as image, speech, and video detection. Additionally, we intend to develop an open-source demonstration system capable of distinguishing between AI-generated content and human-produced content.

Limitations

Despite providing a comprehensive dataset that spans multiple languages, generators, and domains across three distinct tasks in machine-generated text detection, our study encounters several limitations that pave the way for future research.

Firstly, the reliance on textual data without access to white-box information, such as token-level probabilities, confines our detection methods to

black-box approaches across all tasks. These methods might exhibit reduced effectiveness and struggle to generalize across new domains, generators, and languages. Additionally, they are susceptible to language-style attacks, including paraphrasing in different tones, back-translation, and other forms of textual adversarial tactics. In contrast, methods that leverage watermarking and white-box patterns show greater promise for robust MGT detection.

Secondly, our approach to boundary identification presupposes that each text comprises an initial segment written by humans followed by machine-generated content, with only one transition point. However, real-world scenarios often present more complex challenges. It is crucial not only to ascertain the presence of mixed text but also to identify all transition points. Texts may originate from human authors and undergo refinement via machine assistance, or vice versa, encompassing machine generation followed by human revision. Addressing these nuanced scenarios will be a focus of our future research efforts.

Ethics and Broader Impact

This section outlines potential ethical considerations related to our work.

Data Collection and Licenses Our study utilizes pre-existing corpora, specifically the M4 and OUTFOX datasets, which have been publicly released for research purposes under clear licensing agreements.

Security Implications The dataset underpinning our shared task aims to foster the development of robust MGT detection systems. These systems are crucial for identifying and mitigating misuse scenarios, such as curbing the proliferation of automated misinformation campaigns and protecting individuals and institutions from potential financial losses. In fields such as journalism, academia, and legal proceedings, where the authenticity of information is of utmost importance, MGT detection plays a vital role in maintaining content integrity and trust. Furthermore, by enhancing public awareness of the capabilities and limitations of LLMs, we can cultivate a healthy skepticism towards digital content. Effective MGT detection mechanisms are essential for ensuring that users can place their trust in content generated by LLMs.

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References

- Pranjal Aggarwal and Deepanshu Sachdeva. 2024. [Cunlp at semeval-2024 task 8: Classify human and ai generated text](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 1–6, Mexico City, Mexico. Association for Computational Linguistics.
- Huseyin Alecakir, Puja Chakraborty, Pontus Henningsson, Matthijs van Hofslot, and Alon Scheuer. 2024. [Groningen team a at semeval-2024 task 8: Human/machine authorship attribution using a combination of probabilistic and linguistic features](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 1931–1937, Mexico City, Mexico. Association for Computational Linguistics.
- Jainit Bafna, Hardik Mittal, Suyash Sethia, Manish Shrivastava, and Radhika Mamidi. 2024. [Mast kalandar at semeval-2024 task 8: On the trail of textual origins: Roberta-bilstm approach to detect ai-generated text](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 1638–1644, Mexico City, Mexico. Association for Computational Linguistics.
- Sankalp Bahad, Yash Bhaskar, and Parameswari Krishnamurthy. 2024. [Fine-tuning language models for ai vs human generated text detection](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 905–908, Mexico City, Mexico. Association for Computational Linguistics.
- Maha Ben-Fares, Urchade Zaratiana, Simon Hernandez, and Pierre Holat. 2024. [Fi group at semeval-2024 task 8: A syntactically motivated architecture for multilingual machine-generated text detection](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 1155–1160, Mexico City, Mexico. Association for Computational Linguistics.
- Thijs Brekhof, Xuanyi Liu, Yuwen Zhou, and Joris Ruitenbeek. 2024. [Groningen team d at semeval-2024 task 8: Exploring data generation and a combined model for fine-tuning llms for multidomain machine-generated text detection](#). In *Proceedings of*

- the 18th International Workshop on Semantic Evaluation (SemEval-2024), pages 378–385, Mexico City, Mexico. Association for Computational Linguistics.
- Lujia Cao, Ece Lara Kilic, and Katharina Will. 2024. [Kathlalu at semeval-2024 task 8: A comparative analysis of binary classification methods for distinguishing between human and machine-generated text](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 386–389, Mexico City, Mexico. Association for Computational Linguistics.
- Huixin Chen, Jan Büssing, David Rügamer, and Ercong Nie. 2024. [Team mgtd4adl at semeval-2024 task 8: Leveraging \(sentence\) transformer models with contrastive learning for identifying machine-generated text](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 1722–1729, Mexico City, Mexico. Association for Computational Linguistics.
- Vittorio Ciccarelli, Cornelia Genz, Nele Mastracchio, Wiebke Petersen, Anna Stein, and Hanxin Xia. 2024. [Team art-nat-hhu at semeval-2024 task 8: Stylistically informed fusion model for mgt-detection](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 1701–1708, Mexico City, Mexico. Association for Computational Linguistics.
- Alexis Conneau, Kartikay Khandelwal, Naman Goyal, Vishrav Chaudhary, Guillaume Wenzek, Francisco Guzmán, Edouard Grave, Myle Ott, Luke Zettlemoyer, and Veselin Stoyanov. 2019. Unsupervised cross-lingual representation learning at scale. *arXiv preprint arXiv:1911.02116*.
- Evan Crothers, Nathalie Japkowicz, Herna Viktor, and Paula Branco. 2022. Adversarial robustness of neural-statistical features in detection of generative transformers. In *2022 International Joint Conference on Neural Networks (IJCNN)*, pages 1–8. IEEE.
- Evan Crothers, Nathalie Japkowicz, and Herna L Viktor. 2023. Machine-generated text: A comprehensive survey of threat models and detection methods. *IEEE Access*.
- Patrick Darwinkel, Sijbren van Vaals, Marieke van der Holt, and Jarno van Houten. 2024. [Groningen group e at semeval-2024 task 8: Detecting machine-generated texts through pre-trained language models augmented with explicit linguistic-stylistic features](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 995–1003, Mexico City, Mexico. Association for Computational Linguistics.
- Ayan Datta, Aryan Chandramania, and Radhika Mamidi. 2024. [Weighted layer averaging roberta for black-box machine-generated text detection](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 1634–1637, Mexico City, Mexico. Association for Computational Linguistics.
- Rina Donker, Björn Overbeek, Dennis van Thulden, and Oscar Zwagers. 2024. [Groningen team f at semeval-2024 task 8: Detecting machine-generated text using feature-based machine learning models](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 1924–1930, Mexico City, Mexico. Association for Computational Linguistics.
- Liam Dugan, Daphne Ippolito, Arun Kirubakaran, Sherry Shi, and Chris Callison-Burch. 2023. Real or fake text?: Investigating human ability to detect boundaries between human-written and machine-generated text. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 37, pages 12763–12771.
- Seyedeh Fatemeh Ebrahimi, Karim Akhavan Azari, Amirmasoud Irvani, Arian Qazvini, Pouya Sadeghi, Zeinab Taghavi, and Hossein Sameti. 2024. [Sharif-mgtd at semeval-2024 task 8: A transformer-based approach to detect machine generated text](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 552–559, Mexico City, Mexico. Association for Computational Linguistics.
- Chujie Gao, Dongping Chen, Qihui Zhang, Yue Huang, Yao Wan, and Lichao Sun. 2024. Llm-as-a-coauthor: The challenges of detecting llm-human mixcase. *arXiv preprint arXiv:2401.05952*.
- Sebastian Gehrmann, Hendrik Strobelt, and Alexander Rush. 2019a. [GLTR: Statistical detection and visualization of generated text](#). In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics: System Demonstrations*, pages 111–116, Florence, Italy. Association for Computational Linguistics.
- Sebastian Gehrmann, Hendrik Strobelt, and Alexander M Rush. 2019b. Gltr: Statistical detection and visualization of generated text. *arXiv preprint arXiv:1906.04043*.
- Katy Ilonka Gero, Vivian Liu, and Lydia Chilton. 2022. Sparks: Inspiration for science writing using language models. In *Designing interactive systems conference*, pages 1002–1019.
- Renhua Gu and Xiangfeng Meng. 2024. [Aispace at semeval-2024 task 8: A class-balanced soft-voting system for detecting multi-generator machine-generated text](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)*, pages 1487–1492, Mexico City, Mexico. Association for Computational Linguistics.
- Biyang Guo, Xin Zhang, Ziyuan Wang, Minqi Jiang, Jinran Nie, Yuxuan Ding, Jianwei Yue, and Yupeng Wu. 2023. [How close is chatgpt to human experts? comparison corpus, evaluation, and detection](#). *CoRR*, abs/2301.07597.